

**APEX METAL PROTECTION****Section: 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING****1.1 Product identifier**

Product name : APEX METAL PROTECTION
UFI : S7T6-U0Q8-YA0A-0F3J
Product code : 116631E
Use of the Substance/Mixture : Machine Warewashing Detergent
Substance type: : Mixture

For professional users only.

Product dilution information : 0.1 % - 0.3 %

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Dishwash and rinse aid product; Automatic process
Recommended restrictions on use : Reserved for industrial and professional use.

1.3 Details of the supplier of the safety data sheet

Company : Ecolab a.s
Innsporten 9
Postboks 6440-Etterstad, N-0605 Oslo Norway +47 22 68 18 00
NO-kundeservice@ecolab.com

1.4 Emergency telephone number

Emergency telephone number : +4785295496
+32-(0)3-575-5555 Trans-European
Poison Information Centre : +47 22 59 13 00
telephone number

Date of Compilation/Revision : 18.02.2022
version : 1.0

Section: 2. HAZARDS IDENTIFICATION**2.1 Classification of the substance or mixture**

Classification (REGULATION (EC) No 1272/2008)

Product AS SOLD

Eye irritation, Category 2

H319

Product AT USE DILUTION

APEX METAL PROTECTION

Not a hazardous substance or mixture.

2.2 Label elements**Labelling (REGULATION (EC) No 1272/2008)****Product AS SOLD**

Hazard pictograms



Signal Word

: Warning

Hazard Statements

: H319

Causes serious eye irritation.

Hazardous components which must be listed on the label:

Sodium Carbonate

Sodium silicate

Product AT USE DILUTION

Not a hazardous substance or mixture.

2.3 Other hazards**Product AS SOLD**

None known.

Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS**3.2 Mixtures****Product AS SOLD****Hazardous components**

Chemical Name	CAS-No. EC-No. REACH No.	Classification REGULATION (EC) No 1272/2008	Concentration : [%]
Sodium Carbonate	497-19-8 207-838-8 01-2119485498-19	Eye irritation Category 2; H319	≥ 50 - ≤ 100
Sodium silicate	1344-09-8 215-687-4 01-2119448725-31	Skin corrosion Category 1B; H314 Serious eye damage Category 1; H318 Specific target organ toxicity - single exposure Category 3; H335 Serious eye damage/eye irritation Category 1 ≥ 28 % Serious eye damage/eye irritation Category 2A 24 - < 28 % Skin corrosion/irritation Category 1B ≥ 39 % Skin corrosion/irritation Category 2 24 - < 39 % Specific target organ toxicity - single exposure Category 3 ≥ 24 %	≥ 10 - < 20

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alcohols, c13-15-branched and linear, butoxylated ethoxylated	111905-53-4 POLYMER	Skin irritation Category 2; H315 Eye irritation Category 2; H319	$\geq 2.5 - < 5$
oxirane, methyl-, polymer with oxirane	9003-11-6 POLYMER	Acute toxicity Category 4; H332	$\geq 1 - < 2.5$

Product AT USE DILUTION

Remarks : No hazardous ingredients

For the full text of the H-Statements mentioned in this Section, see Section 16.

Section: 4. FIRST AID MEASURES**4.1 Description of first aid measures****Product AS SOLD**

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention.

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

Product AT USE DILUTION

In case of eye contact : Rinse with plenty of water.

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

4.2 Most important symptoms and effects, both acute and delayed

See Section 11 for more detailed information on health effects and symptoms.

4.3 Indication of immediate medical attention and special treatment needed

Treatment : Treat symptomatically.

Section: 5. FIREFIGHTING MEASURES**Product AS SOLD****5.1 Extinguishing media**

Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : None known.

5.2 Special hazards arising from the substance or mixture

Specific hazards during : Not flammable or combustible.

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firefighting

Hazardous combustion products : Depending on combustion properties, decomposition products may include following materials:
Carbon oxides
nitrogen oxides (NOx)
metal oxides

5.3 Advice for firefighters

Special protective equipment for firefighters : Use personal protective equipment.

Further information : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Section: 6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Product AS SOLD

Advice for non-emergency personnel : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Advice for emergency responders : If specialised clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials.

Product AT USE DILUTION

Advice for non-emergency personnel : Refer to protective measures listed in sections 7 and 8.

Advice for emergency responders : If specialised clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials.

6.2 Environmental precautions

Product AS SOLD

Environmental precautions : Do not allow contact with soil, surface or ground water.

Product AT USE DILUTION

Environmental precautions : No special environmental precautions required.

6.3 Methods and materials for containment and cleaning up

Product AS SOLD

Methods for cleaning up : Sweep up and shovel into suitable containers for disposal.

Product AT USE DILUTION

Methods for cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

6.4 Reference to other sections

APEX METAL PROTECTION

See Section 1 for emergency contact information.
For personal protection see section 8.
See Section 13 for additional waste treatment information.

Section: 7. HANDLING AND STORAGE

7.1 Precautions for safe handling

Product AS SOLD

- Advice on safe handling : Avoid contact with skin and eyes. Use only with adequate ventilation. Wash hands thoroughly after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).
- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling.

Product AT USE DILUTION

- Advice on safe handling : Wash hands after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE). For personal protection see section 8.
- Hygiene measures : Wash hands before breaks and immediately after handling the product.

7.2 Conditions for safe storage, including any incompatibilities

Product AS SOLD

- Requirements for storage areas and containers : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.
- Storage temperature : 0 °C to 50 °C

Product AT USE DILUTION

- Requirements for storage areas and containers : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

7.3 Specific end uses

Product AS SOLD

- Specific use(s) : Dishwash and rinse aid product; Automatic process

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Product AS SOLD

Contains no substances with occupational exposure limit values.

DNEL

Sodium Carbonate	:	End Use: Workers Exposure routes: Inhalation Potential health effects: Long-term local effects Value: 10 mg/m3
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		End Use: Consumers Exposure routes: Inhalation Potential health effects: Acute local effects Value: 10 mg/m ³
Sodium silicate	:	End Use: Workers Exposure routes: Inhalation Potential health effects: Long-term systemic effects Value: 5.61 mg/m ³ End Use: Workers Exposure routes: Dermal Potential health effects: Long-term systemic effects Value: 1.59 mg/cm ² End Use: Consumers Exposure routes: Inhalation Potential health effects: Long-term systemic effects Value: 1.38 mg/m ³ End Use: Consumers Exposure routes: Dermal Potential health effects: Long-term systemic effects Value: 0.8 mg/cm ² End Use: Consumers Exposure routes: Ingestion Potential health effects: Long-term systemic effects Value: 0.8 ppm

PNEC

Sodium silicate	:	Fresh water Value: 7.5 mg/l Marine water Value: 1 mg/l Intermittent use/release Value: 7.5 mg/l Sewage treatment plant Value: 348 mg/l
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8.2 Exposure controls**Product AS SOLD****Appropriate engineering controls**

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Individual protection measures

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Hygiene measures	: Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling.
Eye/face protection (EN 166)	: Due to the form and packaging of the product, no protective equipment is needed under normal use conditions.
Hand protection (EN 374)	: No special protective equipment required.
Skin and body protection (EN 14605)	: No special protective equipment required.
Respiratory protection (EN 143, 14387)	: Due to the form and packaging of the product, no protective equipment is needed under normal use conditions.

Product AT USE DILUTION

Appropriate engineering controls

Engineering measures	: Good general ventilation should be sufficient to control worker exposure to airborne contaminants.
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Individual protection measures

Hygiene measures	: Wash hands before breaks and immediately after handling the product.
Eye/face protection (EN 166)	: No special protective equipment required.
Hand protection (EN 374)	: No special protective equipment required.
Skin and body protection (EN 14605)	: No special protective equipment required.
Respiratory protection (EN 143, 14387)	: None required if airborne concentrations are maintained below the exposure limit listed in Exposure Limit Information. Use certified respiratory protection equipment meeting EU requirements(89/656/EEC, (EU) 2016/425), or equivalent, when respiratory risks cannot be avoided or sufficiently limited by technical means of collective protection or by measures, methods or procedures of work organization.

Environmental exposure controls

General advice	: Consider the provision of containment around storage vessels.
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Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

	Product AS SOLD	Product AT USE DILUTION
Physical state	: solid	liquid
Colour	: opaque, light yellow	light yellow
Odour	: slight	slight
pH	: 10.74 - 10.83, 1 %	10.74 - 10.83

APEX METAL PROTECTION

Particle characteristics

Assessment	: no data available	not applicable
Particle size	: no data available	not applicable
Particle Size Distribution	: no data available	not applicable
Dustiness	: no data available	not applicable
Specific surface area	: no data available	not applicable
Surface charge/Zeta potential	: no data available	not applicable
Shape	: no data available	not applicable
Crystallinity	: no data available	not applicable
Surface treatment /Coatings	: no data available	not applicable
Flash point	: Not applicable.	
Odour Threshold	: Not applicable and/or not determined for the mixture	
Melting point/freezing point	: Not applicable and/or not determined for the mixture	
Boiling point, initial boiling point and boiling range	: > 100 °C	
Evaporation rate	: Not applicable and/or not determined for the mixture	
Flammability	: Not applicable and/or not determined for the mixture	
Upper explosion limit	: Not applicable and/or not determined for the mixture	
Lower explosion limit	: Not applicable and/or not determined for the mixture	
Vapour pressure	: Not applicable and/or not determined for the mixture	
Relative vapour density	: Not applicable and/or not determined for the mixture	
Density and / or relative density	: 1.001 - 1.003	
Water solubility	: soluble	
Solubility in other solvents	: Not applicable and/or not determined for the mixture	
Partition coefficient: n-octanol/water (log value)	: Not applicable and/or not determined for the mixture	
Auto-ignition temperature	: Not applicable and/or not determined for the mixture	
Thermal decomposition	: Not applicable and/or not determined for the mixture	
Viscosity, kinematic	: Not applicable and/or not determined for the mixture	
Explosive properties	: Not applicable and/or not determined for the mixture	
Oxidizing properties	: The substance or mixture is not classified as oxidizing.	

9.2 Other information

Not applicable and/or not determined for the mixture

Section: 10. STABILITY AND REACTIVITY**Product AS SOLD****10.1 Reactivity**

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No dangerous reaction known under conditions of normal use.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

No dangerous reaction known under conditions of normal use.

10.4 Conditions to avoid

None known.

10.5 Incompatible materials

Acids

10.6 Hazardous decomposition products

Depending on combustion properties, decomposition products may include following materials:

Carbon oxides

nitrogen oxides (NO_x)

metal oxides

Section: 11. TOXICOLOGICAL INFORMATION

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product AS SOLD

Information on likely routes of exposure : Eye contact, Skin contact

Product

Acute oral toxicity : There is no data available for this product.

Acute inhalation toxicity : 4 h Acute toxicity estimate : > 5 mg/l
Test atmosphere: dust/mist

Acute dermal toxicity : There is no data available for this product.

Skin corrosion/irritation : Mild skin irritation

Serious eye damage/eye irritation : Eye irritation

Respiratory or skin sensitization : Contains an ingredient that can cause asthmatic-like reactions in sulfite-sensitive individuals.

Carcinogenicity : There is no data available for this product.

Reproductive effects : There is no data available for this product.

Germ cell mutagenicity : There is no data available for this product.

Teratogenicity : There is no data available for this product.

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STOT - single exposure : There is no data available for this product.

STOT - repeated exposure : There is no data available for this product.

Aspiration toxicity : There is no data available for this product.

Components

Acute oral toxicity : Sodium Carbonate LD50 rat: 2,800 mg/kg

Sodium silicate LD50 rat: 3,400 mg/kg

alcohols, c13-15-branched and linear, butoxylated ethoxylated
LD50 rat: > 2,000 mg/kg

Components

Acute inhalation toxicity : oxirane, methyl-, polymer with oxirane 4 h LD50 rat: 1 mg/l
Test atmosphere: dust/mist

Components

Acute dermal toxicity : Sodium silicate LD50 rat: > 5,000 mg/kg
Test substance: Information given is based on data obtained from
similar substances.

alcohols, c13-15-branched and linear, butoxylated ethoxylated
LD50 rat: > 2,000 mg/kg

Potential Health Effects

Product AS SOLD

Eyes : Causes serious eye irritation.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes : Health injuries are not known or expected under normal use.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact : Redness, Pain, Irritation

Skin contact : No symptoms known or expected.

APEX METAL PROTECTION

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

Product AT USE DILUTION

Eye contact : No symptoms known or expected.

Skin contact : No symptoms known or expected.

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

11.2 Information on other hazards

Further information : no data available

Section: 12. ECOLOGICAL INFORMATION**12.1 Toxicity****Product AS SOLD**

Environmental Effects : This product has no known ecotoxicological effects.

Product AT USE DILUTION

Environmental Effects : This product has no known ecotoxicological effects.

**Product AS SOLD
Product**

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

Components

Toxicity to fish : Sodium Carbonate^{96 h} LC₅₀ *Lepomis macrochirus* (Bluegill sunfish): 300 mg/l

Sodium silicate^{96 h} LC₅₀ *Oncorhynchus mykiss* (rainbow trout): 260 mg/l

alcohols, c13-15-branched and linear, butoxylated ethoxylated^{96 h} LC₅₀ Fish: 5 mg/l

oxirane, methyl-, polymer with oxirane^{96 h} LC₅₀ Fish: > 100 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : Sodium Carbonate^{48 h} EC₅₀ *Ceriodaphnia* (water flea): 213.5 mg/l

Sodium silicate^{48 h} EC₅₀ *Daphnia magna* (Water flea): 1,700 mg/l

Components

Toxicity to algae : Sodium silicate^{72 h} EC₅₀ *Desmodesmus subspicatus* (green algae): 207 mg/l

APEX METAL PROTECTION

12.2 Persistence and degradability

Product

no data available

Components

Biodegradability : Sodium Carbonate Result: Not applicable - inorganic

Sodium silicate Result: Not applicable - inorganic

alcohols, c13-15-branched and linear, butoxylated
ethoxylated Result: Readily biodegradable. Result: Readily
biodegradable.

oxirane, methyl-, polymer with oxirane Result: Readily
biodegradable.

12.3 Bioaccumulative potential

no data available

12.4 Mobility in soil

no data available

12.5 Results of PBT and vPvB assessment

Product

Assessment : This substance/mixture contains no components considered to be
either persistent, bioaccumulative and toxic (PBT), or very
persistent and very bioaccumulative (vPvB) at levels of 0.1% or
higher.

12.6 Endocrine disrupting properties

The substance/mixture does not contain components considered to have endocrine disrupting
properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100
or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher

12.7 Other adverse effects

no data available

Section: 13. DISPOSAL CONSIDERATIONS

Dispose of in accordance with the European Directives on waste and hazardous waste. Waste
codes should be assigned by the user, preferably in discussion with the waste disposal authorities.

13.1 Waste treatment methods

Product AS SOLD

Product : Where possible recycling is preferred to disposal or incineration. If
recycling is not practicable, dispose of contents/container in
accordance with local regulations Dispose of wastes in an
approved waste disposal facility.

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- Contaminated packaging : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.
- Guidance for Waste Code selection : Inorganic wastes containing dangerous substances. If this product is used in any further processes, the final user must redefine and assign the most appropriate European Waste Catalogue Code. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable European (EU Directive 2008/98/EC) and local regulations.

Product AT USE DILUTION

- Product : Diluted product can be flushed to sanitary sewer if regulations permit.
- Contaminated packaging : Dispose of in accordance with local, state, and federal regulations.

Section: 14. TRANSPORT INFORMATION**Product AS SOLD**

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (ADR/ADN/RID)

- 14.1 UN number or ID number : Not dangerous goods
- 14.2 UN proper shipping name : Not dangerous goods
- 14.3 Transport hazard class(es) : Not dangerous goods
- 14.4 Packing group : Not dangerous goods
- 14.5 Environmental hazards : Not dangerous goods
- 14.6 Special precautions for user : Not dangerous goods

Air transport (IATA)

- 14.1 UN number or ID number : Not dangerous goods
- 14.2 UN proper shipping name : Not dangerous goods
- 14.3 Transport hazard class(es) : Not dangerous goods
- 14.4 Packing group : Not dangerous goods
- 14.5 Environmental hazards : Not dangerous goods
- 14.6 Special precautions for user : Not dangerous goods

Sea transport (IMDG/IMO)

- 14.1 UN number or ID : Not dangerous goods

APEX METAL PROTECTION

number
 14.2 UN proper shipping : Not dangerous goods
 name
 14.3 Transport hazard : Not dangerous goods
 class(es)
 14.4 Packing group : Not dangerous goods
 14.5 Environmental hazards : Not dangerous goods
 14.6 Special precautions for : Not dangerous goods
 user
 14.7 Maritime transport in : Not dangerous goods
 bulk according to IMO
 instruments

Section: 15. REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture
 Seveso III: Directive : Not applicable.
 2012/18/EU of the European
 Parliament and of the Council
 on the control of major-
 accident hazards involving
 dangerous substances.

National Regulations

Take note of Dir 94/33/EC on the protection of young people at work.

Registration number : 603379
 Other regulations : Health and Safety at Work Act.

15.2 Chemical Safety Assessment

No Chemical Safety Assessment has been carried out on the product.

Section: 16. OTHER INFORMATION**Procedure used to derive the classification according to REGULATION (EC) No 1272/2008**

Classification	Justification
Eye irritation 2, H319	Based on product data or assessment

Full text of H-Statements

H314 Causes severe skin burns and eye damage.
 H315 Causes skin irritation.
 H318 Causes serious eye damage.
 H319 Causes serious eye irritation.
 H332 Harmful if inhaled.
 H335 May cause respiratory irritation.

Full text of other abbreviations

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - European Agreement concerning the International Carriage of Dangerous Goods by Road; AIIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada);

APEX METAL PROTECTION

ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

Prepared by : Regulatory Affairs

Numbers quoted in the MSDS are given in the format: 1,000,000 = 1 million and 1,000 = 1 thousand. 0.1 = 1 tenth and 0.001 = 1 thousandth

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

Annex: Exposure Scenarios**Exposure Scenario: Dishwash and rinse aid product; Automatic process**

Life Cycle Stage : Widespread use by professional workers

Product category : **PC35** Washing and cleaning products (including solvent based products)

Contributing scenario controlling environmental exposure for:

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Environmental release category : **ERC8a** Wide dispersive indoor use of processing aids in open systems

Daily amount per site : 7.5 kg

Type of Sewage Treatment Plant : Municipal sewage treatment plant

Contributing scenario controlling worker exposure for:

Process category : **PROC8a** Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Exposure duration : 60 min

Operational conditions and risk management measures : Indoor

Local Exhaust Ventilation is not required

General ventilation Ventilation rate per hour 1

Skin Protection : see section 8

Respiratory Protection : see section 8

Contributing scenario controlling worker exposure for:

Process category : **PROC3** Use in closed batch process (synthesis or formulation)

Exposure duration : 480 min

Operational conditions and risk management measures : Indoor

Local Exhaust Ventilation is not required

General ventilation Ventilation rate per hour 1

Skin Protection : see section 8

Respiratory Protection : see section 8